Issues :-

1) Scraped Data only from 1st page of Croma
2) Duplication of data getting scraped
3) Values of rating fetched have unnecessarily long decimal point values (e.g. 3.222222222)
4) Scrapped non-tv products data

Fixes for issues :-

1) Scraped Data only from 1st page of Croma :-

   - Croma website uses ‘Load More’ pagination technique rather than traditional pagination of [1,2,3] without url parameters
   - Hence this caused scraper to scrap data only from 1st page of product catalog

Fix:-

- Added a function which automatically clicks view-more button until all products are loaded which uses a playwright library

```
def load_all_products(page):
    """Clicks the 'View More' button until all products are loaded."""
    click_count = 0
    while True:
        try:
            view_more_btn = page.locator("button.btn-view-more,
button:has-text('View More')").first

            if view_more_btn.is_visible(timeout=3000):
                view_more_btn.scroll_into_view_if_needed()
                view_more_btn.click()
                click_count += 1
                print(f"Clicked 'View More' {click_count} times...")
                page.wait_for_timeout(4000)
            else:
                print("End of catalog reached.")
                break
        except Exception:
            print("Finished clicking or button became unclickable.")
            break
```

2) Duplicate data getting scraped :-

Handled easily with with this:-

```
unique_tvs = []

seen_names = set()

# Process and deduplicate data
for raw_tv in raw_data:
    tv_info = parse_tv_data(raw_tv)

    if tv_info['Product_Name'] not in seen_names:
        unique_tvs.append(tv_info)
        seen_names.add(tv_info['Product_Name'])
```

3) Values of rating fetched have unnecessarily long decimal point values (e.g. 3.222222222) :-

Altered rating column type to:-

ALTER TABLE led_tv
ALTER COLUMN rating TYPE NUMERIC(3,2);

4) Scrapped non-tv products data :-

Ran these SQL commands to clean :-

- Delete
  from led_tv
  where brand='RD Plast'

- DELETE
  FROM led_tv
  where product_name not like '%LED%' AND product_name not LIKE '%TV%'
  AND description not like '%LED%' AND description not LIKE '%TV%'

- Delete
  from led_tv
  where product_name LIKE '%Stand%'

- Delete
  from led_tv
  where product_name LIKE '%Projector%'

- DELETE
  from led_tv
  where product_name LIKE '%Streaming%'

- DELETE
  from led_tv
  WHERE product_name LIKE '%Mount%'

- DELETE
  from led_tv
  WHERE product_name LIKE '%Smart Remote%'

Learning:-

1) Pagination + optimization by selecting limited data at once :-

   Its efficient to fetch limited amount of data at once rather than entire table , for each page we select limited records with help of **"LIMIT X OFFSET X"**

   Derivation :-

   Consider total_items = 382 (Total records in DB)

   per_page = 12 (Showing 12 records in single page)

   total_pages = ceil(total_items/per_page) = 32 (We will have total 32 pages)

   //offset formula
   offset = (page - 1) * per_page
   (page is current page number , per_page set at default 12)

   For page 3

   offset = (3-1) * 12 = 24

   While fetching data with limit we set LIMIT and OFFSET as

   LIMIT per_page OFFSET offset (offset gets updated dynamically as per page number)

Tools :-

- HTML
- CSS
- JavaScript

- Python/Flask
- Neon DB
- Playwright (for scraping)

References :-

https://www.youtube.com/watch?v=1IK_SSm4NKE&t=553s

https://gemini.google.com/app?hl=en-IN

https://www.youtube.com/watch?v=z1OM9xQEv3A